

Haolan Xu

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EDUCATION

University of Florida

Master of Science in Computer Science

Sept. 2022 - May 2024

Advisor: Prof. Jorg Peters

Sichuan University

Bachelor of Engineering in Chemical Engineering and Technology

Sept. 2018 - Jun. 2022

RESEARCH INTEREST

4D computer vision for **digital humans**: modeling how the human body is shaped, moves, and interacts with the physical world. Currently focused on generating expressive human motion directly from **audio**.

PUBLICATIONS

*: equal contribution

Chun Feng*, Hao Zhang*, **Haolan Xu**, Narendra Ahuja “Seeing is Simulating: Differentiable Physics for Interaction-Aware Material Estimation”, *TMLR*, 2026. [PDF]

Haolan Xu, Keli Cheng, Lei Wang, Ning Bi, Xiaoming Liu “EmoTaG: Emotion-Aware Talking Head Synthesis on Gaussian Splatting with Few-Shot Personalization”, *CVPR*, 2026. [PDF, project page]

Hao Zhang*, **Haolan Xu***, Chun Feng, Varun Jampani, Narendra Ahuja “PhysRig: Differentiable Physics-Based Skinning and Rigging Framework for Realistic Articulated Object Modeling”, *ICCV*, 2025. [PDF, project page]

RESEARCH EXPERIENCE

University of North Carolina at Chapel Hill

Research Assistant in Computer Science

Sept. 2026 - Present

Advisor: Prof. Xiaoming Liu

Michigan State University

Visiting Scholar in Computer Science and Engineering

May 2025 - Aug. 2026

Causal Streaming Audio-Driven 3D Head Animation

Under Review

- Proposed **FlowTalker**, a causal streaming framework for audio-driven 3D head animation, enabling low-latency full-head motion generation for real-time photorealistic head avatar animation.
- Designed a Draft-then-Refine pipeline with spatially factorized motion representations, a kinematics-aware AR transformer, and a selective rectified-flow refiner, achieving coherent full-head motion and fine-grained facial dynamics under causal constraints.

Few-Shot Emotion-Aware 3D Talking Head Synthesis

CVPR 2026

- Proposed **EmoTaG**, a Pretrain-and-Adaptation framework based on 3D Gaussian Splatting, enabling efficient 5-second personalization for audio-driven 3D talking heads with emotion-coherent facial motion.
- Designed a Gated Residual Motion Network (GRMN) guided by a Semantic Emotion Guidance (SEG) module to disentangle linguistic and emotional cues from audio, achieving precise lip articulation and natural facial emotion.

University of Illinois Urbana-Champaign

Research Assistant in Electrical and Computer Engineering

Sept. 2024 - Apr. 2025

Advisor: Prof. Narendra Ahuja

Differentiable Physics-Based 4D Modeling

ICCV 2025 · TMLR 2026

- Proposed **PhysRig**, a differentiable physics-based rigging framework that integrates volumetric soft-body simulation with articulated skeletons for physically grounded and realistic motion synthesis.
- Designed a unified differentiable MPM pipeline that connects particle-level elasticity learning with high-level skeletal control, supporting end-to-end optimization for material-aware dynamic generation.
- Extended the framework to **PhysInteract**, which infers material properties and interaction dynamics from real-world videos for physics-grounded motion understanding.

SELECTED PROJECTS

🔊 Denoise in Real-Time Ray Tracing

🔊 Precomputed Radiance Transfer with Spherical Harmonics

SKILLS SUMMARY

Programming: Python, C/C++

Tools: PyTorch(3D), Blender, GIT, Huggingface, L^AT_EX

Language: English (fluent), Mandarin (native)